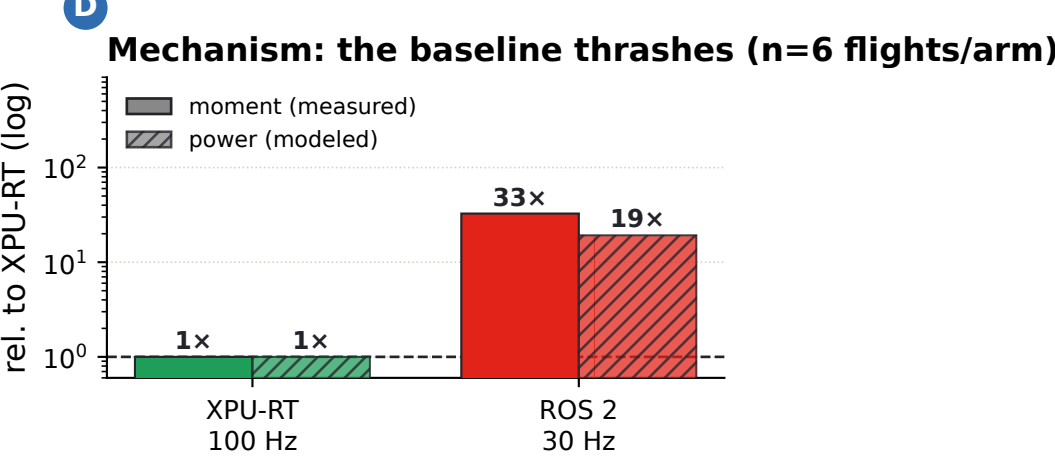
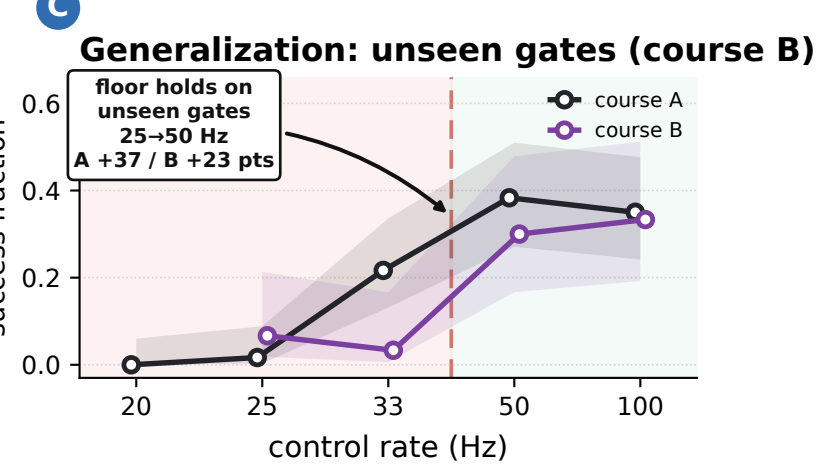
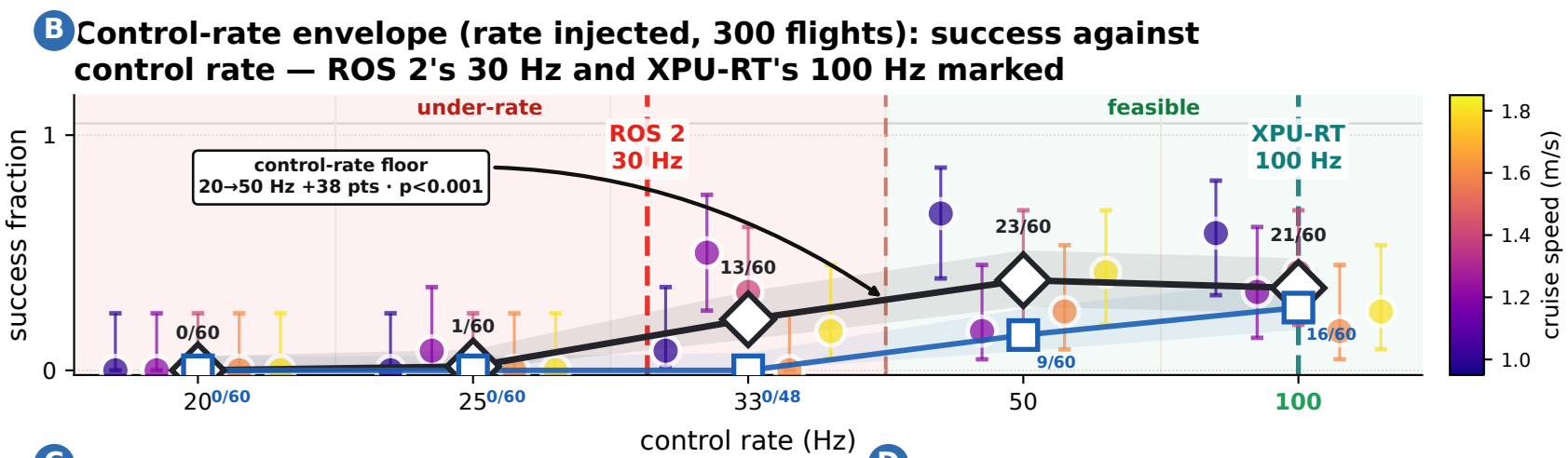
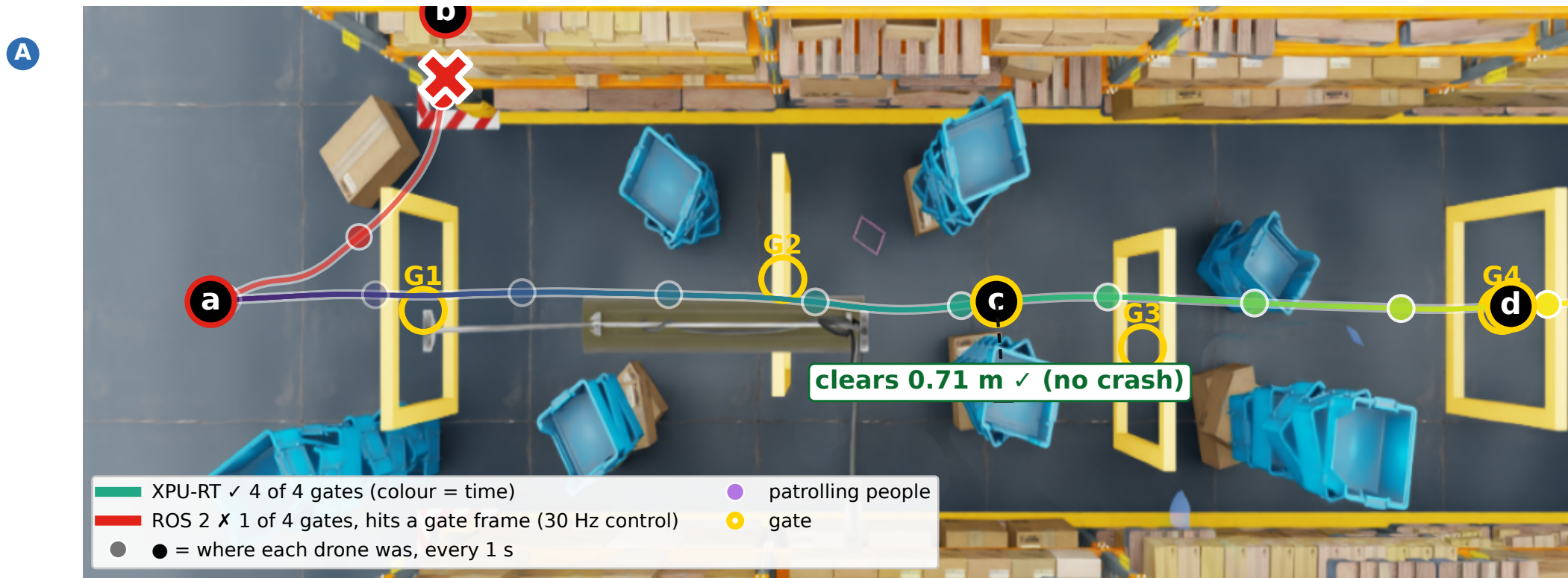
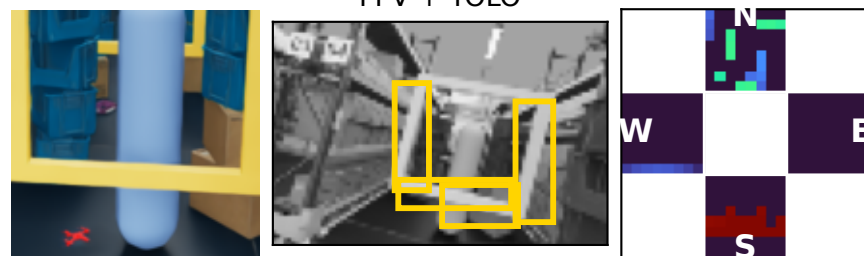


Warehouse gate-course showdown — same scene, drone and gain, at 1.6 m/s
XPU-RT · solver-placed CP-SAT schedule (100 Hz control) clears all 4 gates
ROS 2 · static 6-core partition, control chained to the goal (30 Hz control) clears G1 → hits a gate frame
one seed of this scene (seed 1000); over its 12 seeds XPU-RT completes 5, ROS 2 none



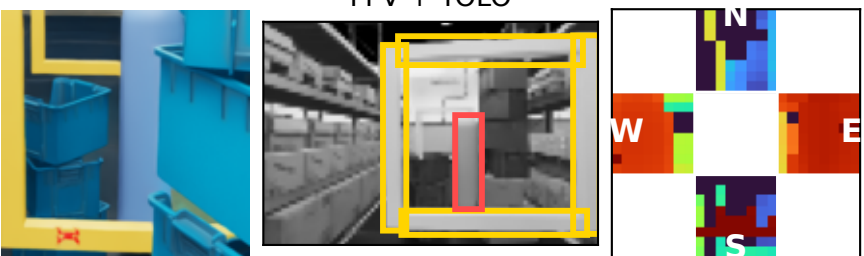
a ROS 2 30 Hz · has not reached G1 · $t=0.7s$
chase



b ROS 2 30 Hz · hits gate frame after G1 · $t=3.0s$
chase



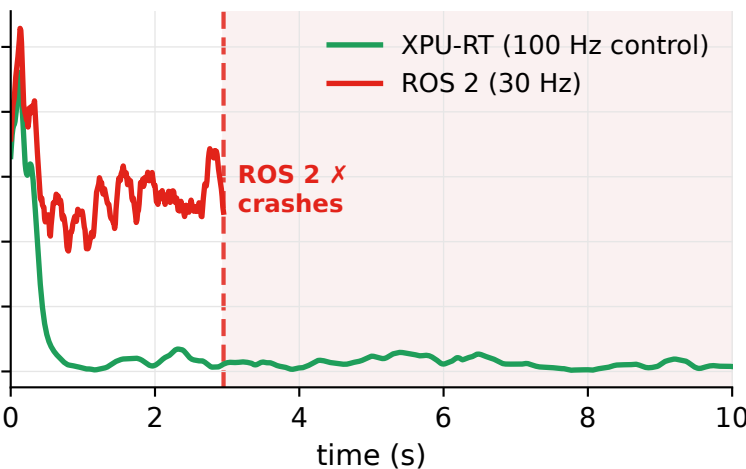
c XPU-RT 100 Hz · clears crate 0.71 m · $t=6.1s$
chase



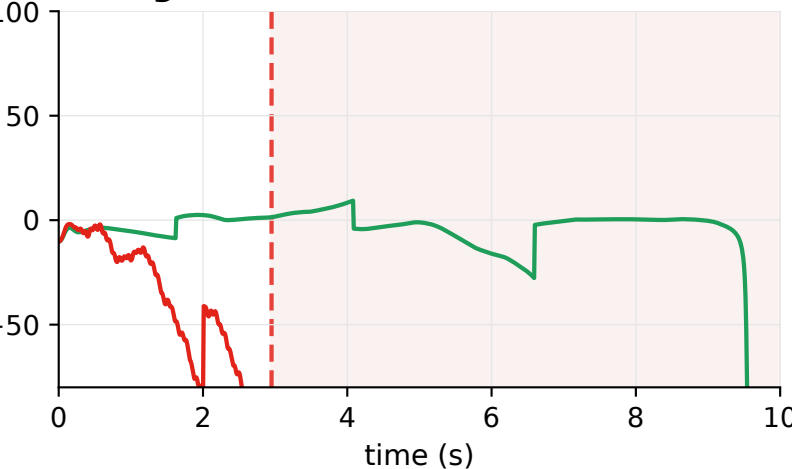
d XPU-RT 100 Hz · reaches gate · $t=9.6s$
chase



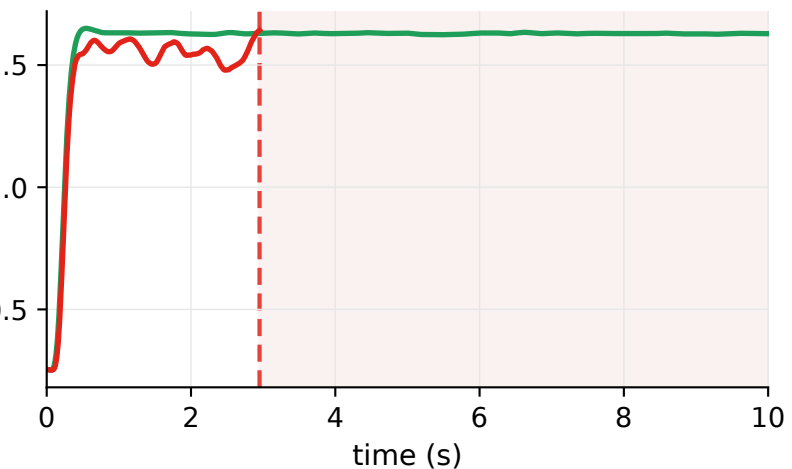
E Inside the pair — body rate: the baseline thrashes at 30 Hz



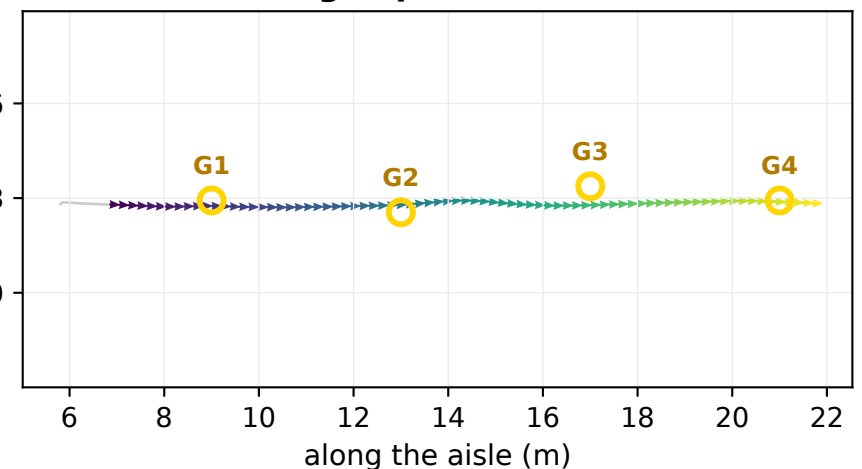
F Nav goal heading: the two commands diverge before the baseline's collision



G Forward speed: the same cruise until the baseline's collision



H XPU-RT's velocity through the gates (arrow = heading · speed, colour = time)



I Onboard K1 schedule, measured at the 30 Hz camera — XPU-RT · solver-placed CP-SAT schedule places the kernels over all 8 harts: camera→control 27 ms, control every 10.0 ms, every frame on time · YOLO on 8 harts, 5 traced, nav on 8 harts, 3 traced, control on 1 hart
ROS 2 timer (3 processes · single executor · pinned): camera→control 34 ms, control every 10.0 ms, 0/599 frames late · YOLO on 4 harts, 1 traced, control on 1 hart, nav on 1 hart
ROS 2 chained (3 processes · single executor · chained control · pinned): camera→control 31 ms, control every 33.4 ms, 0/598 frames late · YOLO on 4 harts, nav on 1 hart, control on 1 hart

